

22593

23124

3 Hours / 70 Marks

Seat No.

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Instructions :

- (1) All Questions are *compulsory*.
- (2) Illustrate your answers with neat sketches wherever necessary.
- (3) Figures to the right indicate full marks.
- (4) Assume suitable data, if necessary.
- (5) Mobile Phone, Pager and any other Electronic Communication devices are not permissible in Examination Hall.

Marks**1. Attempt any FIVE of the following :****10**

- | | | |
|-----|--|---|
| (a) | Give applications of AI. | 2 |
| | Ans: 1. Healthcare, 2. Finance, 3. Transportation, 4. Education | |
| (b) | Define Heuristic Search techniques. | 2 |
| | Ans: A heuristic search is a search strategy that uses heuristic functions to estimate how close a given state n is to the goal. The heuristic helps prioritize which paths to explore first, reducing the overall search space and time. Denoted as $h(n)$. | |
| (c) | List the different types of AI agents. | 2 |
| | Ans: Simple Reflex Agents, Model-Based Reflex Agents, Goal-Based Agents, Utility-Based Agents, and Learning Agents. | |
| (d) | Define Multiclass Classification. | 2 |
| | Ans: Multiclass Classification is a type of machine learning problem where an instance is categorized into one of three or more possible classes or categories. | |
| (e) | List different techniques of data cleaning. | 2 |
| | Ans: Handling missing values, Dealing with outliers, Removing duplicates, correcting inconsistent data. | |
| (f) | Define First – order logic. | 2 |
| | Ans: First-Order Logic (FOL) is a formal system used to represent and reason about statements involving objects, their properties, and relationships, where quantifiers allow for generalization or specification over those objects. It extends propositional logic by incorporating predicates, functions, constants, and variables to express more complex logical propositions. | |

- (g) State any four important supervised machine learning algorithms. **2**
Ans: Linear Regression, Logistic Regression, Decision Trees, and Support Vector Machines (SVM).

2. Attempt any THREE of the following : 12

- (a) Differentiate between Natural (human) Intelligence and Artificial Intelligence. **4**
Ans:

Aspect	Natural Intelligence (Human)	Artificial Intelligence (Machine)
Learning Ability	Learns from experience, senses, and emotions	Learns from data and predefined rules
Creativity	Highly creative and imaginative	Limited creativity; follows programmed logic
Emotions	Capable of emotions, empathy, and moral judgment	Lacks emotions and moral understanding
Processing Speed	Slower (biological processing)	Much faster (electronic processing)
Multitasking	Limited multitasking capability	Can perform multiple tasks simultaneously
Lifespan	Limited (biological limits)	Can operate continuously (until hardware or software fails)
Error Handling	Can make irrational or emotional errors	More consistent, but can make logical errors if misprogrammed

- (b) Discuss the properties of A* algorithm. **4**
Ans:

- 1 Completeness: A* always finds a solution if one exists (finite branching, positive costs).
- 2 Optimality: A* finds the least-cost path if the heuristic is admissible (never overestimates). Guaranteed if the heuristic is also consistent.
- 3 Time Complexity: Exponential in the worst case; increases rapidly with the number of possible paths.
- 4 Space Complexity: Also exponential; requires storing all generated and explored node
- 5 Heuristic Dependence: Efficiency depends on the quality of the heuristic (closer to true cost = faster search).
- 6 Optimal Efficiency: No other algorithm using the same heuristic expands fewer nodes than A*.

- (c) Describe the history and evolution of ML. **4**
Ans: 1940s–1950s: Machine Learning began with the idea of artificial neurons. McCulloch and Pitts created the first neural model (1943), Alan Turing proposed the Turing Test (1950), and Rosenblatt introduced the Perceptron (1957).

1960s–1970s: Progress was slow due to limited computing power. Focus shifted to rule-based AI, leading to the first “AI Winter.”

1980s: Interest revived with the backpropagation algorithm and development of decision trees and Bayesian methods.

1990s: Machine Learning became data-driven with algorithms like SVMs and reinforcement learning.

2000s: The rise of the internet and big data improved ML performance with algorithms like Random Forest and Gradient Boosting.

2010s: Deep learning and neural networks revolutionized ML, especially in image and speech recognition.

2020s–Present: Transformer-based models (like GPT and BERT) dominate. ML is now widely used and focuses on ethics and explainability.

(d) Explain the need of data preprocessing. 4

Ans: Data preprocessing is important because raw data is often incomplete, noisy, and inconsistent. It helps to clean and transform data into a suitable format for analysis or machine learning. Without preprocessing, models may give wrong results or take more time to learn.

1. Handling missing values – Prevents errors in analysis.
2. Removing noise – Reduces inaccurate predictions.
3. Data transformation – Converts data into suitable format.
4. Improves model accuracy – Clean data gives better results.

3. Attempt any THREE of the following : 12

(a) Explain Baye’s theorem. 4

Ans: Bayes' Theorem calculates the probability of a hypothesis based on prior knowledge and new evidence. The formula is:

$$P(H|E) = \frac{P(E|H) \cdot P(H)}{P(E)}$$

Where:

P(H|E): Posterior probability (probability of hypothesis given evidence)

P(E|H): Likelihood (probability of evidence given hypothesis).

P(H): Prior probability (initial probability of hypothesis).

P(E): Marginal likelihood (probability of evidence).

Example: Medical Test

P(Disease) = 0.1 (10% of people have the disease).

P(Pos | Disease) = 0.9 (90% chance of a positive test if diseased).

P(Pos | No Disease) = 0.2 (20% false positive rate).

To find the probability that a person has the disease given a positive test result, use Bayes' Theorem:

$$P(Disease|Pos) = \frac{(0.9 \cdot 0.1)}{0.27} \approx 0.33$$

Even with a positive test, there's only a **33%** chance the person actually has the disease, due to the disease's rarity and false positive rate

(b) Differentiate between Data Analytics and Data Science.

4

Ans:

Data Analytics	Data Science
Process of examining data to find useful insights.	Field of using data to build models, predict, and make decisions.
Focuses on describing past and present data.	Focuses on predicting future trends and decision-making.
Uses statistics, reporting, dashboards.	Uses machine learning, AI, advanced analytics.
To analyze data and answer specific questions.	To create models and discover patterns for future predictions.
Reports, visualizations, summaries.	Predictive models, algorithms, insights.
SQL, Excel, basic stats.	Programming (Python, R), ML, statistics.

(c) Differentiate between overfitting and underfitting.

4

Ans:

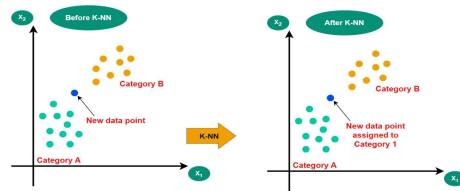
Overfitting	Underfitting
The model learns noise and details too well, performing great on training data but poorly on new data.	The model is too simple and fails to capture underlying patterns in data.
High accuracy on training data, low accuracy on test data.	Poor accuracy on both training and test data.
Too complex a model, too many features, or insufficient data.	Too simple a model or inadequate training.
Use regularization, more data, pruning, or simpler models.	Use more complex models or add features.

(d) Explain any one unsupervised algorithm.

4

Ans: K-Nearest Neighbors (KNN) is primarily a supervised learning algorithm, not unsupervised.

- It's used for classification and regression.
- The algorithm predicts the label of a new data point based on the labels of its **K** closest neighbors in the training data.
- Distance (e.g., Euclidean) is used to find the nearest neighbors.
- The predicted class is usually decided by majority voting (for classification) or averaging (for regression)



4. Attempt any THREE of the following:

12

(a) Describe the structure of agents.

4

Ans: An agent is a system that perceives its environment through sensors and acts upon it through actuators to achieve its goals. The structure of an agent can be divided into four main components:

1 Agent: An agent is a system which perceives the environment and takes actions to achieve its goal.

Example: Robot, self-driving car.

2 Environment: Environment is everything outside the agent which affects its actions.

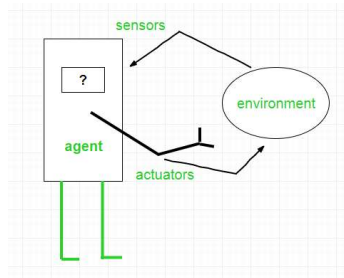
Example: Road, room, web.

3 Sensor: Sensor is a tool used by the agent to observe or measure the environment.

Example: Camera, microphone, temperature sensor.

4 Actuator: Actuator is a tool used by the agent to perform actions in the environment.

Example: Robot's wheels, robotic arm, speaker.



(b) Explain different Heuristic Search techniques.

4

Ans: Heuristic search techniques are used in AI to find solutions quickly when the search space is large. They use heuristic rules (shortcuts or guesses) to select the best path.

Types of Heuristic Search:

1. Hill Climbing: Always moves to the next better state. Simple but can get stuck in local maxima.

Example: Climbing a hill by always going upwards.

2. Best-First Search: Chooses the node that looks closest to the goal using heuristic function.

Example: Finding shortest route in map.

3. A Search: * Uses $f(n) = g(n) + h(n)$, where $g(n)$ = cost so far, $h(n)$ = estimated cost to goal. Gives optimal solution if heuristic is correct.

4. Greedy Search: Always chooses the node nearest to the goal. Fast but may not give the best solution.

(c) Illustrate different types of reasoning in AI. 4

Ans: 1 Deductive Reasoning: This starts from general rules and applies them to a specific case, giving a certain conclusion. Example: All humans are mortal. Ram is a human. Therefore, Ram is mortal.

2 Inductive Reasoning: This starts from specific observations and forms a general rule. The conclusion is only probable. Example: The sun rises every day, so it will rise tomorrow.

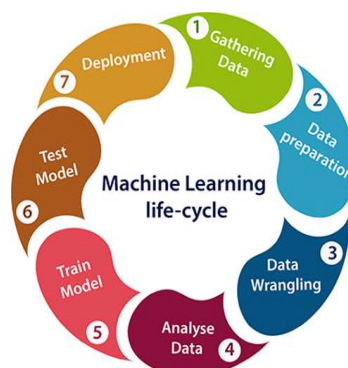
3 Abductive Reasoning: This starts from an observation and finds the most likely explanation. Example: The grass is wet, so it probably rained.

4 Analogical Reasoning: This solves a new problem by comparing it with a similar old problem. Example: A new math problem is solved like a previous similar problem.

(d) Draw and explain the machine learning life cycle. 4

Ans:

1. First, we gather data from different sources like databases, APIs, sensors, or user inputs, because the quality and quantity of the data will determine how well our model can learn.
2. Next, we move on to data preparation, where we clean the raw data by handling missing values, removing duplicates, normalizing numbers, and encoding categories so it's ready for analysis.
3. After that, we do data wrangling, transforming and reshaping the data, creating new features, combining datasets, and filtering out what's not relevant so it's suitable for modeling.
4. Then, we explore the data in data analysis, trying to understand it through statistics and visualizations, checking distributions, correlations, outliers, and trends to get insights that will guide our model.
5. Once the data is ready, we train the model by feeding it into a machine learning algorithm, splitting it into training and validation sets, selecting the right algorithm, and tuning hyperparameters so the model can learn patterns effectively.
6. After training, we test the model on unseen data to see how well it performs, using metrics like accuracy, precision, recall, F1 score, or RMSE, making sure it generalizes well and isn't overfitting.
7. Finally, we deploy the model, integrating it into applications so it can make real-world predictions, and we keep monitoring its performance, updating and retraining it with new data as needed.



(e) State and explain different types of learning.

4

Ans:

Supervised Learning

- Learn from labeled data (input-output pairs).
- The model predicts the output based on input data.
- Examples: Classification, Regression.

Unsupervised Learning

- Learns from unlabeled data (no explicit outputs).
- Finds hidden patterns or groupings in data.
- Examples: Clustering, Dimensionality Reduction.

Semi-Supervised Learning

- Uses a mix of labeled and unlabeled data.
- Useful when labeling data is expensive or limited.

5. Attempt any TWO of the following :

12

(a) Explain Hill Climbing and Best search algorithm with examples.

6

Ans: Hill Climbing is a simple optimization algorithm. Imagine you're standing somewhere on a hilly landscape, blindfolded, and your goal is to reach the highest peak. You can only feel the ground immediately around you, so at each step, you move in the direction that goes uphill the most.

1. Start with an initial solution.
 2. Look at all neighboring solutions.
 3. Move to the neighbor with the highest value (best improvement).
 4. Repeat until no neighbor is better than the current one (you reach a local maximum).
- Limitation: It can get stuck at local maxima or plateaus, where the solution is not the best overall but there's no way to go higher from there.
 - Example: Suppose you are trying to maximize a function ($f(x) = -x^2 + 4x$). You start at ($x = 0$). You evaluate neighbors like ($x = 0.1$) or ($x = -0.1$), and always move to the higher value. Eventually, you'll reach the peak at ($x = 2$), which is the maximum.
 - Imagine adjusting the knobs on a radio to get the clearest signal. You keep turning in the direction that improves the sound, but if there's a slightly better station nearby, you might miss it unless you explore more widely.

Best-First Search (BFS) is a search algorithm that tries to reach the goal by expanding the most "promising" node first. It uses a heuristic to decide which path looks best.

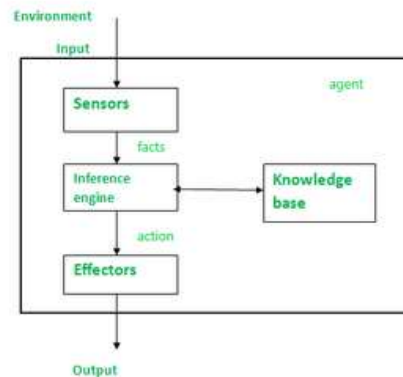
1. Start at the initial node.
 2. Put it in a priority queue based on a heuristic (how close it looks to the goal).
 3. Pick the node with the best heuristic value and expand it (look at its neighbors).
 4. Repeat until the goal is reached.
- Difference from Hill Climbing: While hill climbing only looks at immediate neighbors, best-first search keeps track of all options in a queue and chooses the most promising one overall.
 - Example: Imagine you're trying to get from city A to city B on a map. You always choose the city that seems closest to B based on straight-line distance. You continue this process until you reach B.

- Suppose you're shopping online and trying to find the best deal. You look at all the items, pick the one that seems cheapest or best-rated first, and then explore similar options until you find the best deal.

(b) Describe the architecture and techniques of knowledge based agents in AI.

6

Ans:



A **Knowledge-Based Agent** is a type of intelligent agent that uses knowledge and logical reasoning to make decisions and perform tasks. It differs from simple agents by relying on a structured understanding of its environment, rather than just condition-action rules. This makes it more flexible and capable of handling complex or unfamiliar situations.

The core component of this agent is the **Knowledge Base (KB)**, which stores all the relevant information about the world in the form of facts and rules. This knowledge is usually represented using formal logic, such as propositional or first-order logic, allowing the agent to interpret and reason about its environment systematically.

The **Inference Engine** is another key component that applies logical reasoning to the knowledge base. When new information is received from the environment (via sensors), the inference engine processes it, updates the knowledge base if needed, and determines the best course of action. This reasoning process allows the agent to derive conclusions, make decisions, and solve problems effectively.

(c) Implement multiple linear regression algorithms.

6

Ans:

```

from sklearn.linear_model import LinearRegression
# Feature matrix and target
X = np.array([[1, 2], [2, 3], [4, 5], [3, 2], [5, 4]])
y = np.array([5, 7, 11, 8, 12])
# Create Linear Regression model
model = LinearRegression()
model.fit(X, y)
# Coefficients
print("Intercept:", model.intercept_)
print("Coefficients:", model.coef_)
# Predictions
y_pred = model.predict(X)
print("Predicted values:", y_pred)

```

6. Attempt any TWO of the following :

12

(a) Explain the following Metrics for Regression :

6

- i. Mean Squared Error (MSE)
- ii. Root Mean Squared Error (RMSE)
- iii. Mean Absolute Error (MAE)

Ans:

1. Mean Squared Error (MSE)

Mean Squared Error is the average of the squares of the differences between predicted and actual values.

Formula:

$$\text{MSE} = (1/n) * \sum (y_i - \hat{y}_i)^2$$

Where:

- y_i = actual value
- $\hat{y}_i$ = predicted value
- n = number of observations

Squaring the errors penalizes larger errors more heavily, making MSE sensitive to outliers.

Lower MSE indicates a better-fitting model.

Example:

Actual values: [3, 5, 2]

Predicted values: [2, 5, 4]

$$\text{MSE} = ((3-2)^2 + (5-5)^2 + (2-4)^2) / 3 = (1 + 0 + 4) / 3 = 1.67$$

2. Root Mean Squared Error (RMSE)

RMSE is the square root of MSE.

Formula:

$$\text{RMSE} = \sqrt{\text{MSE}}$$

RMSE has the same units as the target variable, making it easier to interpret.

Like MSE, it penalizes larger errors more heavily.

Example:

$$\text{RMSE} = \sqrt{1.67} \approx 1.29$$

3. Mean Absolute Error (MAE)

MAE is the average of the absolute differences between predicted and actual values.

Formula:

$$\text{MAE} = (1/n) * \sum |y_i - \hat{y}_i|$$

MAE treats all errors equally, making it less sensitive to outliers than MSE. Lower MAE indicates a better model.

Example:

$$\text{MAE} = (|3-2| + |5-5| + |2-4|) / 3 = (1 + 0 + 2) / 3 = 1$$

(b) Explain different forms of data.

6

Ans:

1. Structured Data: Organized in rows and columns. Easy to store and analyze.
Example: Customer info, bank transactions.

2. Unstructured Data: No fixed format; text, images, videos.
Example: Emails, social media posts, photos.
3. Semi-Structured Data: Partially organized with tags or markers.
Example: JSON, XML, HTML, log files.
4. Time-Series Data: Collected over time; shows trends and patterns. Example: Stock prices, sensor readings.
5. Spatial / Geospatial Data: Represents physical locations or coordinates.
Example: GPS coordinates, maps, satellite images.
6. Categorical Data: Represents categories or labels.
Example: Gender, color, ratings.
7. Numerical Data: Quantitative measurements.
Example: Age, salary, temperature.

(c) Elaborate Beyond Classical Search.

6

Ans: Classical search algorithms like BFS, DFS, and uniform cost search work well for simple, fully-defined problems, but they struggle with large, uncertain, or dynamic environments.

To handle real-world problems, AI goes beyond classical search using:

1. Heuristic / Informed Search: Uses smart guesses to find goals faster. Example: A* search.
2. Local Search / Optimization: Improves a single solution iteratively without exploring everything. Example: Hill Climbing, Genetic Algorithms.
3. Adversarial Search: Deals with opponents or competitive environments. Example: Minimax in games.
4. Stochastic / Uncertain Search: Handles randomness or incomplete info. Example: MDPs, POMDPs.
5. Continuous / High-Dimensional Search: Works in continuous or huge spaces. Example: Gradient Descent, Robotics path planning.